

# Fernando Bellido Pazos

Address: Spain | Phone: [+34 638907402](tel:+34638907402) | Email: [fbellidopazos@gmail.com](mailto:fbellidopazos@gmail.com) |  
LinkedIn: [linkedin.com/in/fbellidopazos](https://www.linkedin.com/in/fbellidopazos) | Web: [fbpazos.github.io](https://fbpazos.github.io) | GitHub: [github.com/fbpazos](https://github.com/fbpazos)

## WORK EXPERIENCE

### DXC Technology

AI Engineer & Technical Leader

Zaragoza, Spain  
Jan. 2025 – Today

#### Description:

##### 1. AI Solutions Engineer | GenAI & Architecture

- Led the functional analysis, architecture, and end-to-end development of an enterprise Generative AI pipeline to accelerate technical presales. Replaced manual RFP/tender analysis with a deterministic, schema-first structured extraction and multi-agent auditing system.
- **Architecture & Lifecycle:** Mapped the opportunity lifecycle across 8 stages (Phases 0–7) and designed a modular suite of 9 capabilities to mitigate critical risks, including hidden penalties, tone/style inconsistencies, and bid disqualification caused by envelope-crossover errors.
- **Deterministic (Schema-First) Extraction:** Developed an engine utilizing long-context LLMs and layout-aware document parsing (Docling/PyMuPDF). Bypassed vector RAG in favor of strict, self-healing Pydantic v2 data contracts, structuring 450+ items with end-to-end traceability and verbatim citations.
- **Multi-Agent Evaluation:** Designed a virtual review committee (Neutral Arbiter alongside technical and financial risk evaluators) to audit 150+ page proposals against tender specifications, delivering clause-by-clause verdicts with detailed rationale and textual evidence.
- **Governance & Human-in-the-Loop:** Implemented an inspectable runtime featuring quality gates (Gate M4), enterprise cloud isolation (zero-training data policies), and end-to-end audit logging.
- **Impact & Empirical Validation:** Validated the MVP on live procurement bids (34-page RFP vs. 153-page proposal), slashing compliance matrix generation from 16–47 hours to under 10 minutes (>95% time savings) with 100% coverage across 320+ requirements (under 7% variance).

##### 2. AI Auditing System

- **Led the technical delivery**, managing both developers and a project manager, while also developing the core AI solution.
- Built the agentic system using **Google ADK** to analyze uploaded audit evidences (various file formats and sizes) and determine their sufficiency against SOX controls.
- Generated **detailed, AI-analyzed reports for Control Owners**, providing pre-submission validation that was previously unavailable.
- **Significantly reduced auditor time** for evidence checking, resulting in substantial cost savings by minimizing the required audit resources.
- Empowered Control Owners with a precise **inventory and validation check**, streamlining evidence preparation and improving compliance quality.

##### 3. Enterprise AI Platform Leadership

- Led cross-functional teams to design and launch a **Microsoft AutoGen-powered** multi-agent AI platform, managing various microservices and cloud-native architecture on Azure.
- **Architected intelligent agents** (geospatial, claims, decision automation) at enterprise scale using Azure OpenAI GPT-4o.
- Designed **event-driven data pipelines** (Azure EventHub, 99.9% reliability), processing over 1M records.
- Owned **end-to-end delivery** from technical roadmap to enterprise deployment.
- Built DevOps pipelines, **CI/CD automation**, and **AI model validation** frameworks for scalable production systems.
- **Mentored teams on advanced AI**, multi-agent orchestration, and secure execution environments.

##### 4. Agentic Traffic Management Automation Platform:

- Designed and implemented a multi-agent geospatial planning engine to automate temporary traffic management layouts from high-level roadwork requests, using **Google ADK, Agent2Agent, MCP, PostgreSQL, Azure and Kubernetes**.
- Built an orchestration layer coordinating **specialised agents and shared MCP tools**, ensuring deterministic, auditable workflows and **clean separation between business logic, standards and geospatial computation**.
- Integrated the platform with existing enterprise systems (including CRM and work management tools) via dedicated adapters, enabling **end-to-end automation from request intake to deployment-ready traffic plans**.
- Improved **delivery speed and consistency of traffic management designs** by standardising computations, reducing manual configuration effort and lowering the risk of human error in layout definition.

### IDOM | Consulting, Engineering, Architecture

Consultant & System Innovator

Bilbao, Spain  
Sept. 2023 – Dec. 2024

#### Description:

##### 1. System Innovator:

- Developed a Proof-of-Concept **Digital Twin** utilizing **Autodesk Tandem** and **Azure IoT Hub**, in conjunction with **Azure Stream Analytics** and **Azure ML Studio**, to showcase advanced capabilities. Demonstrated that this approach could outperform traditional human monitoring systems by providing real-time data integration and predictive analytics.
- Directed the theoretical research & development of an innovative internal **Retrieval Augmented Generation (RAG)** system, significantly enhancing language understanding capabilities and applications within the organization. Spearheaded the creation of an AI-driven companion, demonstrating leadership in leveraging advanced AI technologies & Large Language Models

(LLM) to drive organizational progress and innovation.

- Pioneered the development of an **AI classifier using NLP and the Google's BERT** language model, complemented by an ontology-based approach, for sophisticated categorization derived from comprehensive project descriptions.
- Innovated a **CV generator** grounded in the S&P's Success Factors internal system, featuring a multi-layered approach encompassing data ingestion, AI-driven & deterministic data processing paragraph generation, template compilation, and polished CV output.
- Architected and deployed a cutting-edge AI development infrastructure, encapsulating a fully **dockerized** framework that enables real-time monitoring of system resources. Masterfully set-up **Traefik** for efficient routing, and implemented a suite of advanced AI tools, setting new benchmarks in technological excellence like **Stable Diffusion**.

## 2. Consulting – Digital Transformation:

- Developed an **AI-automated pipeline** for processing tabular data into specific SAP naming conventions using **Azure Document Intelligence** and **Azure Phi-3.5** cloud deployment. This system significantly reduced data processing time and improved the accuracy and consistency of data integration within SAP systems compared to manual human processing.
- Spearheaded the development of cutting-edge **artificial intelligence and mathematical models**, revolutionizing asset maintenance from reactive to proactive using S&P's Intelligent Asset Management and Internet of Things modules.
- Played a pivotal role as a Statistician in developing the cadastre system, in collaboration with the registry office, offering clear justifications for property valuations to citizens. Innovated a **unique calculation method for assessing house values** across the Basque Country, enhancing the system's accuracy and transparency in property valuation.

## Ontology Engineering Group

Data Analyst and Developer Intern

Madrid, Spain

Sept. 2021 – Sept. 2022

### Description:

- Developed a sophisticated system to construct **knowledge graphs from academic papers** housed within institutional repositories, enhancing advanced querying capabilities related to Open Science principles.
- Analyzed large datasets of academic publications to extract meaningful metadata and relationships, improving data accessibility and facilitating deeper insights for researchers.
- Implemented **ontology-based data models** to represent complex scholarly information, enabling better interoperability and integration with other data sources.
- **Optimized data ingestion** processes by automating metadata extraction and semantic annotation, increasing efficiency by 40% compared to manual methods.
- Collaborated with a multidisciplinary team to refine algorithms for data processing and knowledge representation, contributing to the advancement of open-source tools in the academic community.
- Utilized technologies such as **RDF, SPARQL**, and Python libraries for **semantic data processing**, enhancing the group's capabilities in semantic web technologies.

## EDUCATION

### Carlos III University of Madrid

MSc in Applied & Computational Mathematics [English]

Madrid, Spain

Sept. 2022 – June 2023

**GPA:** 8.38/10 with 1 Honourable Mention in Advanced Analysis.

**Relevant Coursework:** Biological Inspired Artificial Intelligence, Modelling and Nonlinear Analysis, Optimization, Stochastic Equations for Finance and Biology, Complex Biological and Socioeconomic systems, Computational Linear Algebra, Computational techniques for Differential equations, Methods for Nonlinear differential equations, Applied discrete mathematics.

### Madrid's Polytechnic University

Bachelor's degree in mathematics & computer science [Spanish]

Madrid, Spain

Sept. 2018 – June 2022

**GPA:** 9.4/10 with 14 Honourable mentions.

**Relevant Coursework:** Data Analysis, Software Engineering; Operating Systems, Artificial Intelligence, Databases, Modelling, Programming (Imperative, Object Oriented, Functional, Logic, Systems), IT Project Management, Topology, Calculus, Algebra, Pure Mathematics.

## PROJECTS

### Basic Numerical Methods in Python - Student Oriented

June 2023

**Description:** My end of master's thesis aims to develop an open-source PyPi project that includes a scripted version of the numerical methods taught in universities and a Jupyter-integrated GUI for real-time visualization of those. The goal is to help students understand the methods and promote learning Python instead of MATLAB, which is not as popular or accessible as Python in the current job market.

- This project was awarded the 'Most Innovative Project of 2023' by Logicalis in the Logicalis Innovation Challenge, recognizing its potential to significantly impact educational practices and student learning experiences.

### Sturm Liouville Problem and Models based on differential equations on the field of Acoustics

June 2022

**Description:** Pursued a comprehensive study of the Sturm-Liouville theory and modelled stringed musical instruments to explore the relationship between mathematics and music. The aim of the research was to answer questions such as: what factors contribute to the distinct sound of instruments like a violin and a piano, and how does mathematics play a role in the creation of music.

## AWARDS

- Logicalis Innovation Challenge
- Cambridge Advanced Certificate (C1)
- Academic Excellence Grant 2021 & 2022
- Academic Excellence Award for Representative Agents of Banco Mediolanum 2022

## ADDITIONAL

**People Skills:** Leadership, Public Speaking, Critical Thinking, High Self-Motivation, Dependable  
**Languages:** English (Native) and Spanish (Native)